Marwan Hatem

AI Engineer

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Linked-in | GitHub

Personal Summary

Passionate and driven computer science engineer specializing in artificial intelligence, with academic training in machine learning and deep learning. I have hands-on experience building automation workflows with n8n, integrating them into real-world projects to streamline processes and connect AI-driven components. I combine technical knowledge with strong organizational and leadership skills, and I'm always eager to learn and take on new challenges.

Projects

Desktop AI Office Assistant

Built a fully local employee productivity monitor using webcam-based YOLOv8 detection (person + phone) and Windows active-window classification to track focus, distraction, and away time in real time. Integrated on-device HuggingFace AI — BART-large-CNN for document summarization and MarianMT for multilingual translation — running on CUDA with lazy loading and per-pair model caching. Engineered role-based access with manager approval workflows, a collapsible sidebar UI, persistent session tracking, and a standalone admin panel for full database CRUD and factory reset. Designed with SOLID principles, threaded Qt architecture, and 300+ automated acceptance tests; validated end-to-end on Windows 11 with an RTX 4050 GPU. Tools: Python, PyQt5, YOLOv8, HuggingFace, SQLite

YouTube Content Automation Pipeline (June 2026)

Built an end-to-end automation system using n8n to generate and publish YouTube content, integrating AI-driven script and content generation with automated publishing to LinkedIn. Designed multi-step workflows handling triggers, data transformation, and API integrations across platforms. Tools: n8n, OpenAI/OpenRouter APIs, LinkedIn API

NeuroScan AI – Brain Tumor Classification System (March 2026)

Engineered a custom 5-block Squeeze-and-Excitation Residual CNN from scratch (no transfer learning) to classify brain MRI scans across 15 tumor subtypes plus normal cases, Implemented Grad-CAM explain ability to generate visual heatmaps of the brain regions driving each prediction, and deployed a full-stack diagnostic dashboard with real-time inference, drag-and-drop uploads, an animated confidence gauge, and exportable PDF clinical reports. Tools: PyTorch, FastAPI, Grad-CAM, HTML/CSS/JavaScript

Education

Bachelor of Science in Artificial Intelligence
Arab Open University
September 2023 – February 2027

Courses

Advanced AI Diploma Instant Software Solutions (February 2026 – Present)

Focused on advanced concepts in Artificial Intelligence, including deep learning, computer vision, natural language processing (NLP), large language models (LLMs), Retrieval-Augmented Generation (RAG), and MLOps, with an emphasis on building, deploying, and optimizing real-world AI systems. Worked on an LLM project as part of the program.

Data Engineer & AI – DEPI (July, 2025-Dec, 2025)

Gaining hands-on experience building and optimizing data pipelines, preparing and cleaning datasets for AI applications, and collaborating with ML engineers to support model integration and deployment.

Machine Learning & Computer Vision Fundamentals – Arab Open University (August, 2024)

Focused on foundational concepts in machine learning and image processing, such as object detection, edge detection, image segmentation, and deep learning basics.

Skills

Technical Skills

- Python OOP
- Data Visualization
- Machine learning programing
- Deep learning programing
- Computer Vision
- NLP
- LLMs
- TensorFlow
- PyTorch
- N8N

Personal Skills

- Leadership
- Teamwork
- Presentation skills
- Time Management
- Communication
- Project Management
- Problem Solving